

The Five-Weight Case of a Bernoulli Threshold Conjecture

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Abstract

Let $0 \leq p \leq 1/3$, let $w_1, \dots, w_5$ be positive real numbers with sum one, and let $X_1, \dots, X_5$ be independent Bernoulli(p) random variables. We prove

$$\mathbb{P}\left(\sum_{i=1}^5 w_i X_i \geq p\right) \geq p.$$

In the only nontrivial region, singletons lie below the threshold and all triples lie above it. The graph of nonqualifying pairs has matching number at most one, so it is a star or a triangle. This gives 76 labeled pair threshold families. Their probabilities depend only on the number of qualifying pairs, and two exact polynomial factorizations finish the proof. No assertion is made for a different number of weights or for $p > 1/3$.

1 Introduction

Milojević and Sudakov [1] conjectured that, for arbitrarily many positive weights w_i summing to one and independent Bernoulli(p) variables X_i , the probability that $\sum_i w_i X_i$ is at least its mean p is at least p whenever $0 \leq p \leq 1/3$. They relate the conjecture to a restricted Manickam–Miklós–Singhi problem and obtain a very small unconditional range of p , together with conditional conclusions for larger ranges. The same paper gives counterexamples to the unrestricted statement for $p > 1/3$, apart from $p = 1/2$. Its stated results do not settle the case of five weights throughout $[0, 1/3]$.

This note proves precisely that fixed five-weight case. After elementary containment reductions, the problem becomes a classification of the ten two-element subsets of [5]. The relevant low-pair graphs are exactly the stars and triangles, and their labeled count is 76. Only two probability polynomials need separate sign checks. The argument includes the endpoints $p = 0, 1/5, 1/3$, the faces $w_i = p$, and the pair-threshold faces $w_i + w_j = p$.

The combinatorial input is thus a five-vertex classification of pairwise-intersecting 2-uniform families, in the general Erdős–Ko–Rado setting of intersecting uniform set systems. For adjacent higher-uniform context, specifically a sharp minimum-triple-degree theorem for intersecting 4-uniform families, see [2]. That work is not used here and does not concern weighted Bernoulli sums.

2 Low-pair graphs

For $S \subseteq [5]$, write $w(S) = \sum_{i \in S} w_i$.

Theorem 1. *For every $p \in [0, 1/3]$, every five positive real numbers $w_1, \dots, w_5$ satisfying $\sum_{i=1}^5 w_i = 1$, and mutually independent $X_i \sim \text{Bernoulli}(p)$,*

$$\mathbb{P}\left(\sum_{i=1}^5 w_i X_i \geq p\right) \geq p.$$

The case $p = 0$ is immediate. If $w_i \geq p$ for some i , then $\{X_i = 1\}$ is contained in the threshold event, so the desired probability is at least p . Since $\max_i w_i \geq 1/5$, this also handles all $0 < p \leq 1/5$. It remains to consider

$$\frac{1}{5} < p \leq \frac{1}{3}, \quad 0 < w_i < p \quad (1 \leq i \leq 5), \quad \sum_{i=1}^5 w_i = 1. \quad (1)$$

In this region no singleton qualifies. Every triple S does qualify, because

$$w(S) = 1 - w(S^c) > 1 - 2p \geq p.$$

Thus every set of cardinality at least three qualifies, and only pairs can vary.

Put $x_i = w_i/p$ and $R = \sum_i x_i = 1/p$. Then

$$0 < x_i < 1, \quad R \geq 3. \quad (2)$$

Let L be the graph on $[5]$ in which ij is an edge exactly when the pair is nonqualifying:

$$ij \in E(L) \iff x_i + x_j < 1.$$

The strict inequality is essential: a pair with total weight exactly p qualifies in the original non-strict threshold event.

Lemma 2. *The graph L is a star (possibly with zero edges) or a triangle together with two isolated vertices. Conversely, every such labeled graph occurs for some parameters satisfying (1). Hence there are exactly 76 labeled low-pair graphs, and every admissible family has at least six qualifying pairs.*

Proof. If ij and $k\ell$ were disjoint edges of L , and r were the remaining vertex, then (2) would give

$$R = (x_i + x_j) + (x_k + x_\ell) + x_r < 1 + 1 + 1 = 3,$$

a contradiction. Thus every two edges of L intersect. If an intersecting family of edges is not a star, choose two edges ab, ac and an edge not containing their common vertex a . That edge must be bc ; any further edge would have to meet all three edges of this triangle and hence would itself be one of them. This proves the asserted graph classification.

For completeness, Table 1 gives one exact normalized weight vector for every isomorphism type. The coordinates are assigned by the roles indicated in the middle column. In each row, all coordinates lie strictly between zero and one, $3 \leq R < 5$, and direct addition realizes exactly the displayed strict low edges. Taking $p = 1/R$ and $w_i = x_i/R$ therefore gives (1). The labeled counts follow by choosing the edge, star center and leaves, or triangle vertices. Their sum is

$$1 + 10 + 30 + 20 + 10 + 5 = 76.$$

Since a low graph has at most four edges, at least six of the ten pairs qualify. □

L	assignment of the x_i	R	qualifying pairs q	count
empty	every vertex $3/5$	3	10	1
one edge	endpoints $2/5$; others $4/5$	$16/5$	9	10
two-edge star	center $1/10$; leaves $4/5$; others $19/20$	$18/5$	8	30
three-edge star	center $1/10$; leaves $4/5$; other $19/20$	$69/20$	7	20
triangle	triangle vertices $12/25$; others $4/5$	$76/25$	7	10
four-edge star	center $1/10$; leaves $3/4$	$31/10$	6	5

Table 1: Exact realizations of all low-pair graph types.

We shall also need the following refinement of the last conclusion.

Lemma 3. *If exactly six pairs qualify, then $p > 1/4$.*

Proof. There are then four low edges, so choose one of them, say ij . If r, s, t are the other vertices, then

$$R = (x_i + x_j) + x_r + x_s + x_t < 1 + 1 + 1 + 1 = 4.$$

Since $R = 1/p$, this is equivalent to $p > 1/4$. $\square$

3 Probability polynomials

Let q denote the number of qualifying pairs. In the reduced region (1), the empty set and singletons do not qualify, all ten triples, all five four-element sets, and the full set qualify. Independence therefore gives

$$F_q(p) = qp^2(1-p)^3 + 10p^3(1-p)^2 + 5p^4(1-p) + p^5 \quad (3)$$

for the probability in Theorem 1. Moreover, $F_{q+1}(p) - F_q(p) = p^2(1-p)^3 > 0$ here.

If $q \geq 7$, it is enough to use

$$F_7(p) - p = p(1-p)(p^3 - 5p^2 + 6p - 1). \quad (4)$$

For $g(p) = p^3 - 5p^2 + 6p - 1$,

$$g(1/5) = \frac{1}{125} > 0, \quad g'(p) = 3p^2 - 10p + 6 \geq \frac{8}{3} > 0$$

on $[1/5, 1/3]$. Thus (4) is positive throughout the reduced region.

It remains only to consider $q = 6$. Lemma 3 gives $p > 1/4$, and

$$F_6(p) - p = p(1-p)(-3p^2 + 5p - 1). \quad (5)$$

For $h(p) = -3p^2 + 5p - 1$,

$$h(1/4) = \frac{1}{16} > 0, \quad h'(p) = 5 - 6p \geq 3 > 0$$

on $[1/4, 1/3]$. Hence (5) is positive whenever a six-pair family is feasible. These cases exhaust (1); together with the initial containment cases, they prove Theorem 1.

Computational and AI assistance

The proof above is self-contained. A Python 3 program using only integer and exact rational arithmetic was used to enumerate all 2^{10} labeled graphs, check exact realization witnesses for the 76 admissible patterns, and verify the polynomial identities in (4) and (5). These checks were used as corroboration, not as a substitute for the classification and sign arguments. OpenAI Codex assisted with developing and organizing the proof, checking boundary cases, generating and checking exact code, and preparing the manuscript. Every argument needed for the theorem is included in the text.

References

- [1] Aleksa Milojević and Benny Sudakov. On the Probability a Weighted Bernoulli Sum Exceeds Its Mean, 2026. arXiv:2606.30287 [math.PR].
- [2] Zijian Zeng. A sharp 3-degree erdős–ko–rado theorem for 4-uniform families. SSRN working paper 7379380, 2026.